

A - Supermajority

Time Limit: 2 sec / Memory Limit: 1024 MiB

Score : 100 points

Problem Statement

You are given positive integers A and B .

If $A > B \times \frac{2}{3}$, output Yes; otherwise, output No.

Constraints

- $1 \leq A, B \leq 1000$
 - All input values are integers.
-

Input

The input is given from Standard Input in the following format:

$A \ B$

Output

Output the answer.

Sample Input 1

316 465

Sample Output 1

Yes

Since $316 > 465 \times \frac{2}{3}$, output Yes.

Sample Input 2

101 248

Sample Output 2

No

Since $101 > 248 \times \frac{2}{3}$ does not hold, output No.

Sample Input 3

666 999

Sample Output 3

No

Since $666 > 999 \times \frac{2}{3}$ does not hold, output No.

B - Parking 2

Time Limit: 2 sec / Memory Limit: 1024 MiB

Score : 200 points

Problem Statement

There is a parking lot. The fee for parking in this lot is as follows:

- During the period from exactly L o'clock to exactly R o'clock, a fee of X is charged for each hour parked.
- During the period not covered above, a fee of Y is charged for each hour parked.

If a car is parked in this lot from exactly A o'clock to exactly B o'clock without crossing midnight, how much is the fee?

Constraints

- $1 \leq X, Y \leq 1000$
- $1 \leq L < R \leq 23$
- $1 \leq A < B \leq 23$
- All input values are integers.

Input

The input is given from Standard Input in the following format:

```
 $X$   $Y$   $L$   $R$   $A$   $B$ 
```

Output

Output the answer.

Sample Input 1

```
700 300 9 17 7 21
```

Sample Output 1

```
7400
```

If a car is parked from 7 o'clock to 21 o'clock, the fee is as follows.

- For the 2 hours from 7 o'clock to 9 o'clock, a fee of $300 \times 2 = 600$ is charged.
- For the 8 hours from 9 o'clock to 17 o'clock, a fee of $700 \times 8 = 5600$ is charged.
- For the 4 hours from 17 o'clock to 21 o'clock, a fee of $300 \times 4 = 1200$ is charged.

The answer is $600 + 5600 + 1200 = 7400$.

Sample Input 2

```
600 500 9 17 17 20
```

Sample Output 2

```
1500
```

If a car is parked from 17 o'clock to 20 o'clock, the fee is as follows.

- For the 3 hours from 17 o'clock to 20 o'clock, a fee of $500 \times 3 = 1500$ is charged.

The answer is 1500.

Sample Input 3

```
900 200 12 14 11 13
```

Sample Output 3

```
1100
```

C - Reverse Permutation

Time Limit: 2 sec / Memory Limit: 1024 MiB

Score : 300 points

Problem Statement

You are given an integer N and a string S of length N consisting of o and x.

There is an integer sequence $A = (A_1, A_2, \dots, A_N)$ of length N . Initially, $A = (1, 2, \dots, N)$.

Perform the following operation on A for $k = 1, 2, \dots, N$ in this order.

- If $S_k = \text{o}$, reverse the first k terms of A . Specifically, replace A with $(A_k, A_{k-1}, \dots, A_1, A_{k+1}, A_{k+2}, \dots, A_N)$.
- If $S_k = \text{x}$, do nothing.

Find A after all the operations are completed.

Constraints

- $2 \leq N \leq 5 \times 10^5$
- N is an integer.
- S is a string of length N consisting of o and x.

Input

The input is given from Standard Input in the following format:

```
 $N$   
 $S$ 
```

Output

Output the elements of A after all the operations are completed, separated by spaces.

Sample Input 1

```
5  
ooxoo
```

Sample Output 1

```
5 2 1 3 4
```

A changes as follows with each operation:

- For $k = 1$: reverse the first 1 term of A . A becomes $(1, 2, 3, 4, 5)$.
- For $k = 2$: reverse the first 2 terms of A . A becomes $(2, 1, 3, 4, 5)$.
- For $k = 3$: do nothing.
- For $k = 4$: reverse the first 4 terms of A . A becomes $(4, 3, 1, 2, 5)$.
- For $k = 5$: reverse the first 5 terms of A . A becomes $(5, 2, 1, 3, 4)$.

After all the operations are completed, A is $A = (5, 2, 1, 3, 4)$.

Sample Input 2

```
7  
0000000
```

Sample Output 2

```
7 5 3 1 2 4 6
```

Sample Input 3

```
15  
X00X0X0X0X0XX00
```

Sample Output 3

```
15 11 10 7 6 3 1 2 4 5 8 9 12 13 14
```

D - X to Y

Time Limit: 2 sec / Memory Limit: 1024 MiB

Score : 400 points

Problem Statement

You are given integers X , Y , and an integer K that is at least 2.

There is a variable x , which is initially $x = X$. You can perform the following operation on x zero or more times:

- Choose an integer y satisfying $\left\lfloor \frac{x}{K} \right\rfloor = y$ or $\left\lfloor \frac{y}{K} \right\rfloor = x$, and replace the value of x with y .

Here, $\lfloor z \rfloor$ is defined as the greatest integer not exceeding z , for a real number z .

Find the minimum number of operations required to make $x = Y$. Under the given constraints, it can be proved that there always exists a way to make $x = Y$ in a finite number of operations.

You are given T test cases; solve each of them.

Constraints

- $1 \leq T \leq 2 \times 10^5$
- $0 \leq X, Y \leq 10^{18}$
- $2 \leq K \leq 10^{18}$
- All input values are integers.

Input

The input is given from Standard Input in the following format:

```
T
case1
case2
⋮
caseT
```

The i -th ($1 \leq i \leq T$) test case case _{i} is given in the following format:

```
X Y K
```

Output

Output the answers for the test cases in order, separated by newlines.

Sample Input 1

```
4
11 9 3
0 0 2
842 180 7
1948706013487601 48019760148910476 89014537
```

Sample Output 1

```
2
0
7
5
```

Consider the first test case.

By performing the following operations, $x = Y$ can be achieved in two operations:

- Choose $y = 3$. Since $\left\lfloor \frac{x}{K} \right\rfloor = \left\lfloor \frac{11}{3} \right\rfloor = 3$, this choice is valid. Then, replace the value of x with 3.
- Choose $y = 9$. Since $\left\lfloor \frac{y}{K} \right\rfloor = \left\lfloor \frac{9}{3} \right\rfloor = 3$, this choice is valid. Then, replace the value of x with 9.

$x = Y$ cannot be achieved in fewer than two operations, so output 2 on the first line.

E - Digit Circus

Time Limit: 2 sec / Memory Limit: 1024 MiB

Score : 450 points

Problem Statement

Find, modulo 998244353, the number of integers x with $1 \leq x \leq N$ that satisfy **exactly one** of the following three conditions.

- x is a multiple of 3.
- The decimal representation of x contains 3.
- Exactly three different digits are used in the decimal representation of x .

Here, the decimal representation of an integer should not have unnecessary leading 0s.

Constraints

- N is an integer.
- $1 \leq N < 10^{500}$

Input

The input is given from Standard Input in the following format:

N

Output

Output the answer.

Sample Input 1

45

Sample Output 1

19

Among the integers from 1 to 45, there are 19 integers satisfying exactly one of the conditions in the problem statement, as follows.

- There are 10 integers satisfying only the first condition: 6, 9, 12, 15, 18, 21, 24, 27, 42, 45.
- There are 9 integers satisfying only the second condition: 13, 23, 31, 32, 34, 35, 37, 38, 43.
- There are no integers satisfying only the third condition.

Sample Input 2

1013

Sample Output 2

424

Below are examples of applicable integers.

- Integers satisfying only the first condition include 555 and 1011.
- Integers satisfying only the second condition include 343 and 553.
- Integers satisfying only the third condition include 1012 and 704.

Sample Input 3

2

Sample Output 3

0

There are no applicable integers.

Sample Input 4

314159265358979323846264338327950

Sample Output 4

```
658111391
```

F - Sjeltzer?

Time Limit: 2 sec / Memory Limit: 1024 MiB

Score : 525 points

Problem Statement

A certain refrigerator assigns each drink a 6-digit numeric string as its ID.

This refrigerator stores N drinks. The i -th drink has ID S_i and size V_i . $S_1, \dots, S_N$ are distinct.

Answer Q queries. Each query is given in the form of 6-digit numeric strings x, y , and asks the following.

- Among the drinks stored in this refrigerator, find the sum of the sizes of those whose assigned ID s satisfies the following for every $k = 1, \dots, 6$.
 - $x_k \leq s_k \leq y_k$, where s_k, x_k, y_k denote the values represented by the k -th digits of s, x, y , respectively.

Constraints

- N is an integer.
- $1 \leq N \leq 3 \times 10^5$
- S_i is a string consisting of digits (0-9).
- $|S_i| = 6$
- $S_1, \dots, S_N$ are distinct.
- V_i is an integer.
- $1 \leq V_i \leq 10^9$
- Q is an integer.
- $1 \leq Q \leq 3 \times 10^5$
- In each query, x and y are strings consisting of digits (0-9).
- In each query, $|x| = |y| = 6$.

Input

The input is given from Standard Input in the following format:

```

 $N$ 
 $s_1$   $V_1$ 
...
 $s_N$   $V_N$ 
 $Q$ 
query1
⋮
query $Q$ 

```

Here, query _{q} is the q -th query, given in the following format:

```

 $x$   $y$ 

```

Output

Output Q lines. The q -th line should contain the answer to the q -th query.

Sample Input 1

```

5
000000 1
314159 2
161803 10
169231 5
384400 20
4
150001 269944
302010 396939
000000 999999
999000 000444

```

Sample Output 1

```
15
0
38
0
```

For the first query, the drinks satisfying the condition are the third and fourth drinks.

For the second query, no drinks satisfy the condition.

For the third query, the drinks satisfying the condition are the first, second, third, fourth, and fifth drinks.

For the fourth query, no drinks satisfy the condition.

Sample Input 2

```
8
778722 12744734
488742 359266786
955942 335004463
454854 331175626
934246 179761526
927777 657312747
652568 109900930
753279 405123850
8
201000 785589
202325 955898
310401 875947
044023 988999
111230 567897
133241 577989
101033 999499
453013 796988
```

Sample Output 2

```
515024780
109900930
0
331175626
331175626
331175626
584885376
331175626
```

G - Sum of Mex of Mod of Linear

Time Limit: 3 sec / Memory Limit: 1024 MiB

Score : 625 points

Problem Statement

You are given integers N, M, C, K and an integer sequence $A = (A_1, A_2, \dots, A_N)$ of length N .

You are given Q queries, which you should process in order. The q -th ($1 \leq q \leq Q$) query gives integers i_q, X_q , and you should solve the following problem after changing A_{i_q} to X_q .

Find the value of $\sum_{k=0}^{K-1} \text{mex}_{1 \leq i \leq N} \{(Ck + A_i) \bmod M\}$. Here, $\text{mex}_{1 \leq i \leq N} B_i$ denotes the smallest non-negative integer not contained in B , for an integer sequence $B = (B_1, B_2, \dots, B_N)$.

Constraints

- $1 \leq N \leq 2 \times 10^5$
- $0 \leq C < M \leq 10^9$
- $1 \leq K \leq 10^9$
- $0 \leq A_i < M$
- $1 \leq Q \leq 2 \times 10^5$
- $1 \leq i_q \leq N$
- $0 \leq X_q < M$
- All input values are integers.

Input

The input is given from Standard Input in the following format:

```

N M C K
A_1 A_2 ... A_N
Q
i_1 X_1
i_2 X_2
⋮
i_Q X_Q

```

Output

Output Q lines.

The q -th line ($1 \leq q \leq Q$) should contain the answer to the problem after processing the q -th query.

Sample Input 1

```
2 3 1 2
0 2
3
2 2
1 1
2 1
```

Sample Output 1

```
3
1
0
```

The first query sets $A = (0, 2)$. The sought value is $\text{mex}(\{0, 2\}) + \text{mex}(\{1, 0\}) = 1 + 2 = 3$.

The second query sets $A = (1, 2)$. The sought value is $\text{mex}(\{1, 2\}) + \text{mex}(\{2, 0\}) = 0 + 1 = 1$.

The third query sets $A = (1, 1)$. The sought value is $\text{mex}(\{1, 1\}) + \text{mex}(\{2, 2\}) = 0 + 0 = 0$.

Sample Input 2

```
7 9 3 19
7 0 2 8 1 3 4
7
3 5
1 6
3 2
5 5
4 7
7 8
2 4
```

Sample Output 2

```
32
44
53
37
49
37
54
```

Hints: AtCoder Beginner Contest 465

Taken from the official editorial, not written by us. Read one hint at a time.

A. Supermajority *(editorial, first line)*

Find the value $B \times \frac{2}{3}$ as a floating-point number, and compare it with A to find the answer.

B. Parking 2 *(editorial, first line)*

The fee within the h o'clock (from exactly h o'clock to exactly $(h + 1)$ o'clock) can be computed as follows: X if $L \leq h \leq R - 1$; Y otherwise For the integer h , run ...

C. Reverse Permutation *(editorial, first line)*

First, let us rephrase the procedure as follows: Let $A = ()$.; For $k = 1, 2, \dots, N$, do the following: Append k to the end of A .

D. X to Y *(editorial, first line)*

Consider the graph with edges between integers that can be replaced from one to another.

E. Digit Circus *(editorial, first line)*

We can solve the problem using the method called Digit DP (Dynamic Programming).

F. Sjeltzer? *(editorial, first line)*

x In this subproblem, the range of s of interest is those s such that $s_k \leq y_k$ for all k .

G. Sum of Mex of Mod of Linear *(editorial, first line)*

Let c_x be the number of integers k between 0 (inclusive) and K (exclusive) satisfying $Ck + x \equiv 0 \pmod{M}$.

Source: <https://atcoder.jp/contests/abc465/editorial>